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Sounding Odors: What Underlies the Association of Sound and Odor?

ריחות מזמרים: מה המנגנון התומך בקשר גובה צליל-ריח?

Ph.D. Research Proposal

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Abstract

Odors are notoriously difficult to identify and verbalize. Consequently, researchers in chemosensation have increasingly turned to exploring cross-modal correspondences involving odor. In this study, we examined the relationship between odor and pitch. Initial findings indicate systematic associations between specific odors and tone frequencies, suggesting systematic mapping between the two modalities. However, the underlying mechanism of these associations remains unclear. Two possible explanations have been proposed: (a) a low-level neurological basis involving direct sensory interactions, (b) heightened sensitivity to cross-modal features through learned associations, or (c) a high-level cognitive process driven by linguistic or conceptual mappings. Our research aims to further establish the robustness of these associations and then deploy Garner's test of selective attention to support one of the hypotheses.

Vision and olfaction routinely co-occur, and the two systems interact: when colors accompany odors, they can change odor perception, including identification accuracy, response latencies, intensity, and pleasantness (Zellner, 2013). Appropriate color-odor pairings facilitate faster and more accurate identification, whereas incongruent or absent color cues hinder recognition (Zellner, Bartoli, & Eckard, 1991). Notably, liking is higher for odors that people correctly identify, consistent with the role of functional familiarity in olfactory evaluation (Zellner et al., 1991). These basic facts frame our problem that odor is hard to name and identify in isolation, yet it is readily shaped by cross-modal context. Consequently, in the present study we pursued the existence and effects of odor-audition association.

As far back as the fifth century BC, Plato noted, "the varieties of smell have no name" (Cooper & Hutchinson, 1997). People are remarkably poor at identifying odors, with identification rates for even familiar smells often falling significantly below those of other sensory modalities (Algom & Cain, 1991; Algom et al., 1993; Cain & Algom, 1997). Unlike vision, where the Munsell system offers a detailed framework for describing colors along dimensions of hue, value, and chroma, or audition, where sounds are systematically mapped by frequency (pitch) and amplitude (loudness), olfaction lacks a universally accepted dimensional model. This gap highlighted the inherent complexity of odors and hindered advances in both theory and application alike (Di Stefano & Spence, 2023). Keeping these constraints in mind, there is a modest literature on odor-vision associations, whereas the literature is much more sparser on odor-audition association. We studied the latter in this research.

Scientific Study of Olfaction: From Past to Present

Zwaardemaker (1895) pioneered quantitative olfactory measurements through his development of the sliding-cylinder olfactometer. This apparatus enabled precise control over odor stimulus intensity by adjusting the exposed surface area of scent-impregnated tubes. Building on this innovation, Henning (1916) proposed a "prism" model for odors, as described by Marks (1978; see Figure 1). In this model, six primary odors floral, fruity, resinous, spicy, burnt, and putrid were positioned at the vertices of a triangular prism. Odors were conceptualized as points within this three-dimensional structure, representing combinations of the primary qualities based on their relative proximity to the vertices. This geometric framework aimed to systematically map the relationships between different odors, inspired by similar models in color. While innovative, the prism struggled

to account for the full complexity of olfactory perception, highlighting the challenges of defining smell through spatial models.

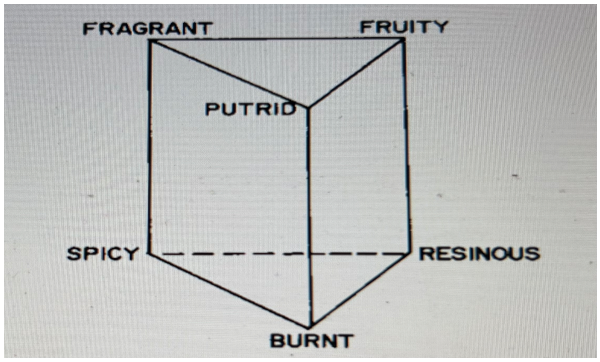

Fig 1: Henning's model as presented in Marks (1978).

In 1971, Schiffman and Erickson introduced a three-dimensional taste model using multidimensional scaling (MDS). They addressed the limitations of Henning's closed-ended structure. Incorporating dimensions such as molecular weight, deviation from neutral pH, and hedonic ratings, their model allowed for a more flexible and precise mapping of taste perception. Building on this, Schiffman (1974) expanded the application of her methodology to olfaction, proposing a two-dimensional model centered on pleasantness and sharpness. This refinement emphasized psychological dimensions and offered an empirical alternative to Henning's odor prism. Subsequent research, including that of Berglund et al. (1973) and Jones et al. (1978), sought to characterize odors through psychophysical dimensions such as intensity and a pleasantness-unpleasantness continuum. These dimensions provided a degree of structure to the study of olfactory perception, although they did not yield a comprehensive framework.

Research into olfactory perception increasingly pointed to pleasantness as a key organizing dimension. Building on earlier work (e.g., Schiffman, 1974), Yeshurun and Sobel (2010) argued that hedonic evaluation is a central function of the olfactory system. They proposed that odors are defined by their pleasantness, a quality that, unlike physical properties, is inherently subjective and resists precise verbalization. Much like an emotion, pleasantness lacks clear linguistic delineation, underscoring the unique challenges of describing and quantifying olfactory experience.

Yeshurun & Sobel (2010) argued that olfactory perception is primarily structured along a hedonic axis rather than by discrete, well-defined perceptual dimensions. Unlike vision or audition, where stimuli can be arranged based on wavelength or frequency, odors resist such classification due to their inherently affective nature. This affective structure suggests that odor correspondences with other sensory modalities including pitch may not

operate through direct sensory integration but rather through mechanisms driven by emotional and hedonic associations. Rather than reflecting direct neural interactions, these correspondences may emerge from a combination of perceptual mechanisms and affective processing. If odor-pitch associations stem from a low-level sensory origin, we would expect them to resemble cross-modal interactions observed in other perceptual domains. However, if these associations are primarily mediated by hedonic processing, their nature may differ significantly, aligning more closely with the way pleasantness and emotional valence structure olfactory perception.

Given that odors are primarily processed through affective and associative mechanisms, it is perhaps unsurprising that they also pose significant challenges for verbalization. Despite the vast range of smells that humans can distinguish, studies consistently show that people struggle to name even familiar odors, with correct identification rates often falling below 50% (Zellner, 2013; Zellner et al., 1991). This difficulty stems from the weak cognitive link between olfactory perception and verbal labels. Phenomena such as the "tip-of-the-nose" effect, where individuals recognize an odor as familiar but cannot name it, further highlight the disconnect between olfactory perception and linguistic representation (Spence, 2020). Additionally, while olfactory memory is robust in sensory terms, it faces significant constraints in translating perceptual data into symbolic or verbal expressions (Algom & Cain, 1991).

Askenazi and Marks (2004) showed that endogenous attention enhances the detectability of gustatory but not olfactory stimuli, suggesting that attention to flavors may be spatially directed toward the tongue, where taste but not smell receptors are located. Consistent with this, visual and contextual cues often compensate for the limitations of olfactory attention, with congruent visual elements such as color enhancing odor recognition and speed. In contrast, incongruent or absent cues hinder identification (Zellner, 1991, 2013).

Cultural factors play a crucial role in shaping olfactory perception, particularly in categorization. For instance, some societies, such as the Jahai in Southeast Asia, possess dedicated abstract terms for smells, which allows for more effective odor categorization (Deroy, 2023). This contrasts sharply with Western urban populations, whose reliance on contextual or visual aids reflects a relative lack of abstraction in their olfactory lexicon. This linguistic and cultural influence is rooted in fundamental cognitive mechanisms: classic work on conceptual metaphor shows that cross-domain mappings are part of everyday cognition, thereby systematically structuring perception (Lakoff, 1993). Furthermore, culture shapes the very form of these mappings, as demonstrated by the

CORE principle, which shows that abstract domains are spatialized according to correlations in experience, leading to culture-specific conceptualizations (Pitt & Casasanto, 2020). Recent cross-cultural analyses further argue that thinking patterns, cognitive frameworks, language practices, and social consensus jointly determine metaphor comprehension across cultures (Zou, Fuller, & Wang, 2025).Cognitive mechanisms, such as inhibitory processes, may also obscure human olfactory capabilities; a study shows improved discrimination under conditions of reduced inhibition, such as after mild alcohol consumption, and the authors link this to the to changes in brain activity in smell zones (Endevelt-Shapira et al., 2014). These findings highlight the complex interaction between sensory perception, cognitive processes, and cultural context in shaping our experience and understanding of odors.

Cross-Modal Research with Odor

Cross-modal sensory experiences often evoke analogous perceptions across different modalities (Marks, 1983). For instance, brightness is associated with loudness, pitch with physical size, red with warmth, and blue with cold. Additionally, the perceived intensity of a stimulus in one sensory modality often aligns with that of another (e.g., Marks, 1983; Melara, 1989; Stevens, 1975). The inherent difficulty in verbalizing odors frequently leads individuals to describe smells using cross-modal references (e.g., "sharp smell"). Such tendencies may reveal an underlying unity in sensory processing. Marks (1978) suggested that these semantic associations reflect a fundamental interconnectedness among the senses. Expanding on this idea, recent work by Di Stefano and Spence (2023) posits that perceptual similarity ranges from phenomenologically driven to conceptually mediated, with variations in cognitive penetrability and associative strength.

A particularly intriguing case of cross-modal correspondence involves odor and pitch. Previous research has documented systematic pitch–odor and harmony–odor associations. For instance, the smell of musk has been linked to brass instruments, whereas those of candied orange and dried plums have been linked to the piano. In general, sweeter-smelling stimuli are associated with higher pitch (Belkin et al., 1997; Crisinel et al., 2012, 2013; Crisinel & Spence, 2012). Interestingly, biological and chemical analyses show that olfactory percepts are not arbitrary, as molecular properties such as molecular weight and the number of carbon and oxygen atoms systematically determine odor bioactivity, suggesting that physical features of molecules may partly

underlie their cross-modal mapping to sound (Boevé & Giot, 2023). Recent studies have shown that such cross-modal associations can affect environmental evaluations and human behavior, including path choice and duration of stay in urban environments (Ba & Kang, 2019a; Ba et al., 2020).

Zacharakis and colleagues (2024, 2025) further investigated odor-sound associations by having musically trained participants systematically match synthesized auditory timbres with aromatic oils, rating their associations' strength. The results indicated consistent pairings; for example, fruity and sour odors were associated with higher pitch and timbres emphasizing high-frequency harmonics, while sweet and spicy scents corresponded to lower pitches and timbres characterized by noise and spectral complexity. Subsequent analysis using shared semantic scales confirmed that semantic attributes such as "bright," "sharp," and "warm" mediated these mappings, reinforcing the idea that conceptual frameworks profoundly shape cross-modal correspondences.

Developing odor-pitch associations as a potential scale for odor classification could provide a breakthrough in our ability to characterize and communicate about odors. This approach is particularly promising given that pitch already serves as a well-established scale for organizing and describing sounds. The question remains, however, about what mechanisms sustain these systematic associations.

Two Hypotheses

Two broad hypotheses have been suggested to account for these odor-pitch correspondences:

1. **The Linguistic Hypothesis.** Metaphors and semantic affiliation sustain the associations, as evinced in common expressions like "sweet child" or "sharp argument". This semantic coding may reflect the fundamental unity of the senses (Marks, 1978). Recent empirical support for this hypothesis is provided by Zacharakis et al. (2024, 2025), who showed that individuals consistently align odors with sounds based on shared semantic dimensions, such as "sharpness," "warmth" and "brightness". These findings suggest that semantic categories guide cross-modal correspondences. They reinforce the central role of conceptual mediation. Psychological-construction accounts propose that words function as abstract features that compress high-dimensional sensory signals and invite perceivers to search for equivalence across instances. In this view, labels such as

sweet or sharp can recruit low-dimensional affective features (e.g., pleasantness) that organize both odor and sound, allowing semantic mediation of odor-pitch mappings (Hoemann & Barrett, 2019; Barrett & Theriault, 2025).

2. **The Sensory-Synesthesia Hypothesis.** The underlying mechanism is based on synesthesia, a condition where individuals (synesthetes) automatically and consistently experience perceptions in one sensory modality in response to stimulation in another (Martino & Marks, 2001). For instance, a synesthete might perceive high-pitched sounds as lighter or sharper visual colors. Importantly, the strength of these experiences often varies: while some report vivid and distracting sensations, others describe them as subtle yet consistent (Marks, 2001). Neural hyperconnectivity between sensory regions provides further evidence for these cross-modal correspondences, supporting both their consistency and intensity (Marks & Mulvenna, 2013). There is also evidence that information integration from two modalities occurs at low-level primary sensory areas (Gilday & Mizrahi, 2023). Recent findings support direct connections between olfactory and auditory brain areas, suggesting a neurobiological basis for these correspondences. Anderson et al. (2003) demonstrated that odor intensity and quality are processed through distinct neural pathways, with amygdala activation linked specifically to intensity. Keller and Vosshall (2004) found that individuals are more consistent in judging odor intensity than pleasantness. Zarzo (2008) showed that odor quality is concentration-dependent and that the hedonic dimension becomes dominant when odors are evaluated at similar intensities. Accordingly, we will obtain trial-wise ratings of pleasantness and intensity and test mediation.

The Present Project

This research is planned to investigate odor-pitch associations and determine their underlying mechanisms. This goal dictates the plan of the study. The first objective is to establish firmly the existence of systematic odor-pitch associations. The second contingent objective is to identify the underlying mechanism for this cross-modal correspondence.

Ethics and Consent

All procedures were approved by the institutional ethics committee and conducted in accordance with the Declaration of Helsinki. Informed consent was obtained, participation

was voluntary, and no personal data were recorded. Participants received modest compensation.

A. Establishing the Presence of Odor-Pitch Associations

Pilot Study A1: Odor-Pitch Associations Among Professional Perfumers

The purpose of Pilot Study A1 was to determine whether professional perfumers, who possess extensive experience with odors, can consistently associate specific odors with particular pitch frequencies. Inspired by Belkin et al. (1997), who demonstrated stable cross-modal correspondences between olfactory and auditory perceptions among non-expert participants, this study explicitly focuses on experts in olfactory perception. Belkin et al. presented participants with odors across two sessions conducted on separate days, requiring them to match each odor with corresponding auditory pitches. Adjustments to pitch were made via the participant's feedback, which resulted in stable odor-pitch associations over time.

Given their specialized training, professional perfumers are hypothesized to demonstrate similar enhanced consistency in odor-pitch mappings. Confirmation of stable associations among perfumers would justify subsequent broader investigations into odor-pitch correspondences. Conversely, if expert perfumers do not show consistent associations, further research in this direction would not be warranted.

Participants

Two professional olfactory evaluators participated in this study. Both are employed by a startup in the olfactory technology industry and have received extensive in-house training in odor identification and evaluation. Evaluator 1 was a 40 year old female. She completed 20 odor-to-pitch matches. Evaluator 2 was a 33 year old male. He produced 45 matches. Both evaluators possess several years of professional experience in perfumery and routine olfactory assessment.

Apparatus and Stimuli

Auditory stimuli were delivered using a Lenovo laptop computer connected to closed-back JBL headphones, minimizing ambient noise interference. Tones were generated via an online digital tone generator (available at www.szynalski.com/tone-generator), allowing precise frequency adjustment (see Figure 2).

Odor stimuli consisted of distinct aromatic compounds, both natural essential oils and synthetic aroma chemicals frequently employed in the fragrance industry. Each odorant was carefully stored in small, amber-colored glass bottles (15 ml), which maintained chemical stability and prevented degradation from light exposure. Standard perfume blotter strips were utilized for precise odor presentation. The number of odor stimuli presented differed between participants, with 20 odorants administered to Participant 1 and 45 to Participant 2 (details of odors in the Appendices).

Design

Each perfumer performed two sessions. Hence, the participants are professional and to avoid variables like memory and learning, subjects used the same large set of odors in two sessions, two weeks apart. Each participant evaluated the odorants by matching each odor with the corresponding auditory pitch frequency. The male perfumer evaluated 20 distinct odorants, and the female perfumer evaluated 40 distinct odorants in the lab. The participants performed the evaluations twice.. Each evaluation session lasted approximately one hour.

The laptop screen displaying numerical frequency values was covered throughout the experiment to prevent visual biases, ensuring participants relied exclusively on auditory-olfactory associations rather than numeric visual cues.

Procedure

The participant was seated comfortably at a desk. For each trial, odor stimuli were individually presented in unmarked amber glass bottles placed on the desk. The bottles contained odorants whose identities were unknown to participants. Lab assistants dipped standard perfume blotter strips into the bottles to deliver the odor. Participant's sniffed each odor directly from the blotter strip near their noses, inhaling naturally while concentrating on the scent qualities; immediately following the sniff, the experimenter activated a continuous auditory tone delivered binaurally through JBL closed-back headphones. The participant then adjusted the frequency via an interactive online tone-generator interface (www.szynalski.com/tone-generator) on the Lenovo laptop, enabling continuous frequency adjustments (Figure 1). Participants were instructed to intuitively

select the auditory pitch frequency that best matched their perceptual experience of each odor.

The online tone-generator interface digitally recorded the participants' pitch selections. No feedback regarding pitch selection accuracy was provided during the trials. Inter-trial intervals ranged from approximately 30 seconds to one minute to minimize adaptation. After completing the first experimental session, the participant returned two weeks later and repeated the procedure with the same odor stimuli, enabling the evaluation of test-retest reliability. The primary analysis assessed the correlation between the pitch selections that each participant made across the two sessions.

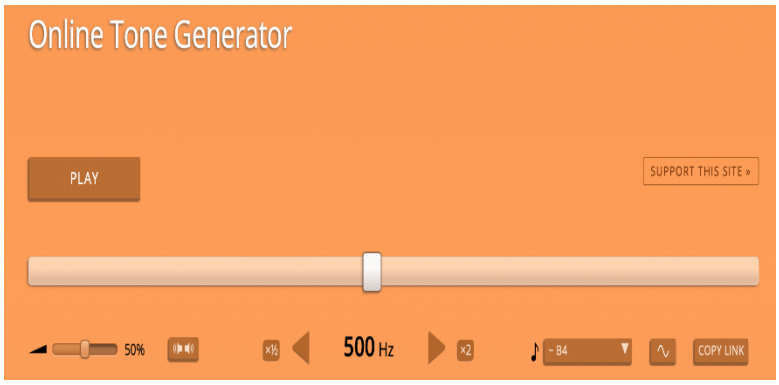

Fig 2: *Illustration of the Online tone Generator interface by Tomasz P. Szynalski (https://www.szynalski.com/tone-generator/) used to present auditory stimuli. The image shows the generator set at 500 Hz. During actual testing, the numerical frequency value was hidden from participants to avoid visual bias.*

Results

The two experts consistently and systematically mapped odors onto auditory pitch. As depicted in Figure 2, the cross-session correlation was r = .79 for Evaluator 1 and r = .66 for Evaluator 2.

The convergence of the conservative (raw) and stringent (outlier-pruned) estimates reinforces the contention, advanced by Spence, Zellner, and Algom, that odor-pitch correspondences are genuine, stable perceptual mappings rather than transient associations. Non-parametric checks (Spearman ρ) confirmed the same ordinal pattern, ruling out the possibility that the improvement is merely a linear-model artifact.

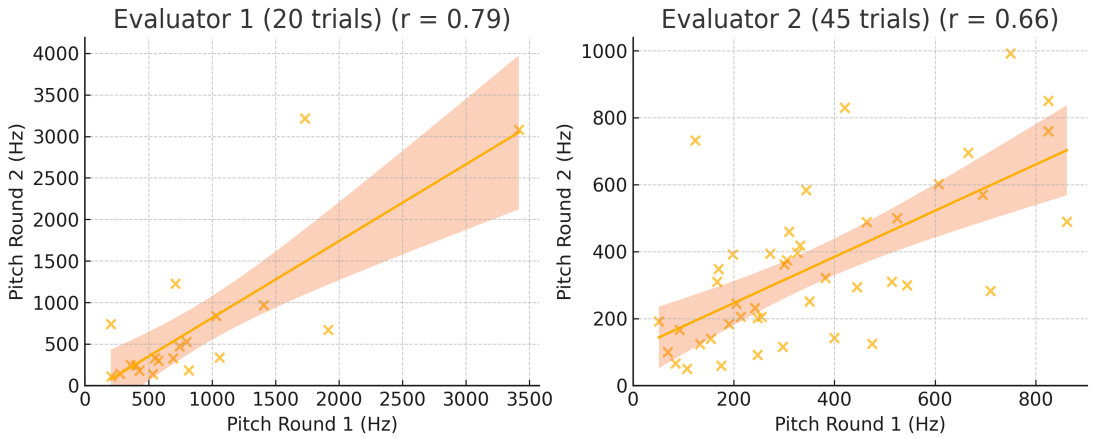

Fig 3: Pitch values assigned to the same odors across two sessions separated by two weeks, plotted for each expert evaluator (left = Evaluator 1, right = Evaluator 2). Each dot represents a single odorant; the red line is the least-squares fit, and the shaded band is the 95 % confidence interval.

Closer examination of the odor-pitch assignments revealed some distinct patterns. Certain odors were consistently assigned t high-pitch values, with Ambermax and Octyl Butyrate receiving pitch assignments above 1700 Hz. These findings suggest that these particular odors may be perceived as sharp or intense, aligning with previous reports of cross-modal correspondences between high-pitched sounds and bright or piercing sensory attributes. In contrast, other odors, such as Eucalyptol and Benzyl Acetate, were systematically assigned lower pitch values, ranging between 200 and 400 Hz. This pattern may reflect their perception as deeper, milder, or smoother odors, consistent with associations between lower-pitched sounds and heavier or warmer perceptual qualities.

Limitations

Although in this domain and particularly with expert evaluators two participants do not necessarily constitute a trivial dataset, the expert sample in Study A1 is still small. Moreover, expertise is a double-edged sword: while it affords highly practiced, internally consistent mappings, it may also limit generalizability to naïve perceivers. Finally, due to technical constraints, the experts and the subsequent naïve sample did not evaluate the exact same odor set. This substantive limitation motivates an immediate test with naïve participants and a fixed, repeatable set of nameless odors.

Second, the two experts used markedly different response scales: Evaluator 1 tended to assign higher pitches (≈500 to 3500 Hz) than Evaluator 2 (≈200 to 800 Hz), indicating

idiosyncratic scaling despite good within-evaluator consistency. Moreover, for Evaluator 1, a single very high assignment (~3500 Hz) exerts disproportionate leverage on the cross-session correlation (r = .79), so this coefficient should be interpreted with caution.

Study A2: Naïve Participants

Method

Participants

Forty adults (mean age ≈ 24; mixed gender) reported normal hearing and olfaction and no formal musical training.

Materials and Apparatus

Seven nameless odors served as stimuli: Eugenol, Citronella, Linalool, Indole, Galaxolide, Pea Rose, and Gallbascone. Odors were presented in identical containers to avoid visual cues. Auditory stimulation followed A1: a continuous pure tone was delivered over headphones from a laptop-based online pitch generator. The numerical frequency readout on the screen was concealed throughout. The starting pitch was 500.

Design

A within-subjects design crossed Odor (7) with Test (three presentations of each odor). The dependent measure was selected frequency (Hz). This design parallels A1 in task logic while using a standardized, repeatable odor set for naïve participants.

Procedure

On each trial, participants smelled one nameless odor and then adjusted the pitch that best matched it. Each odor was presented three times in randomized order, with brief breaks between presentations. Instructions and apparatus were as in A1 except for the shorter within-session breaks (there was no two-week interval), reflecting the goal of estimating stability within a single multi-block session.

Results

Descriptive statistics appear in Table 1 (Summary of Odor -Pitch Associations), and a means plot is provided (Figure: Means of Pitch-Odor, Study A2). The repeated-measures

ANOVA yielded marginal main effects of Test and Odor, with no interaction: Test, $F(2,78)=2.682$, $p=.075$, $\eta^2=.004$; Odor, $F(6,234)=1.842$, $p=.092$, $\eta^2=.026$; Test × Odor, $F(12,468)=1.387$, $p=.168$, $\eta^2=.012$. Means showed a systematic ordering: Indole at the lowest range (~481 Hz), Citronella highest (~745 Hz), with Galaxolide (~712 Hz) and Linalool (~660 Hz) above Eugenol and Pea Rose (~540 Hz each), and Gallbascone intermediate (~579 Hz). Thus, even among naïve participants responding to nameless odors, pitch selections varied in a consistent, odor-specific manner across repetitions, while the absence of a Test × Odor interaction suggests stable relative ordering across the three presentations.

Odor	Mean Pitch (Hz)	Standard Deviation
Eugenol	542.99	327.32
Cirtonelly	745.07	740.86
Linolool	660.05	455.40
Indole	481.02	289.87
Galaxolide	712.30	410.39
Pea rose	540.20	327.34
Gallbascone	579.49	578.06

Table 1: Summary of Odor-Pitch Associations

Helmert Contrast - RM Factor 1

Comparison	Estimate	SE	df	t	p
Indole - Galbascone, PEA, Eugenol, Linalool, Citronell, Galaxolide	-132.577	43.587	228	3.042-	0.003
Galbascone - PEA, Eugenol, Linalool, Citronell, Galaxolide	86.564-	44.205	227.9999	1.958-	0.051
PEA - Eugenol, Linalool, Citronell, Galaxolide	-105.139	45.116	227.9999	2.330-	0.021
Eugenol - Linalool, Citronell, Galaxolide	-125.769	46.596	228	2.699-	0.007
Linalool - Citronell, Galaxolide	46.705-	49.422	228	0.945-	0.346
Citronell - Galaxolide	53.855-	57.068	228	0.944-	0.346

Table 2: Helmet RM Factor

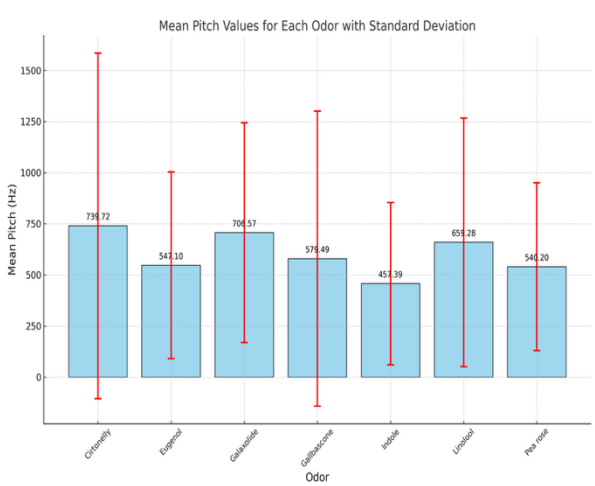

Fig 4: Means of Pitch-Odor.

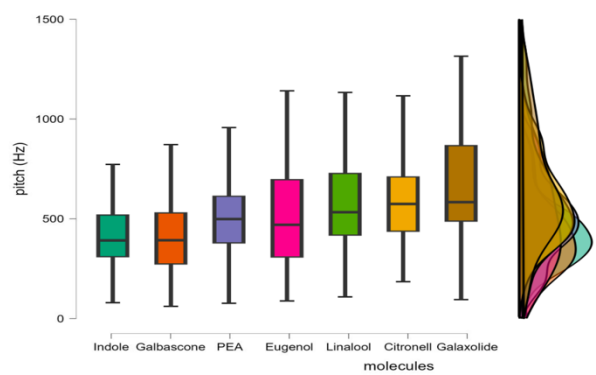

Fig 5: Distribion of Pitch-Odor.

Study A3: Naïve Participants (Four Fixed Choices)

Method

Participants

Twenty-seven adults were tested; one participant who almost always selected the highest category was excluded a priori, yielding N=26 for analysis.

Materials and Apparatus

The same seven nameless odors were used (Eugenol, Citronella, Linalool, Indole, Galaxolide, Pea Rose, Gallbascone). Auditory choices were constrained to four fixed frequencies: 481, 540, 712, and 745 Hz. These values were selected from A2 as the two lowest and two highest odor-means, to test whether the ordinal "places" observed under free adjustment would persist under categorical choice. Apparatus and masking of on-screen numerals matched A1/A2.

Procedure

Across two sessions, each odor was presented twice per session in randomized order. On each trial, participants heard the four pitches (presentation order randomized per trial) and chose the one that best matched the odor. No feedback was given. This procedure mirrors A2's logic while replacing free adjustment with forced choice to reduce noise and probe ordinal stability directly.

Results

Choice distributions showed robust, odor-specific preferences (Figure: Frequency of Pitch Selection, Study A3). Indole was overwhelmingly paired with 481 Hz, and Citronella with 745 Hz; Eugenol and Galaxolide were primarily matched with 540 Hz and 712 Hz, respectively, preserving the A2-based hierarchy under categorical responding. Planned Helmert contrasts confirmed these patterns: Indole vs. others favored lower pitch, $t(156)=-5.81$, $p<.001$; Citronella vs. others favored higher pitch, $t(156)=-2.97$, $p=.003$. Together with the A2 means, these results indicate that odors "retain their place" along a pitch continuum across tasks that differ in response format (free adjustment vs. four-AFC), supporting the conclusion that the mapping is not an artifact of unconstrained tuning.

Cross-Study Note. Studies A2 and A3 converge: when naïve participants match nameless odors to pitch, both free and categorical responses yield the same ordinal configuration

low (e.g., Indole) to high (e.g., Citronella) with intermediate odors (e.g., Eugenol, Galaxolide, Gallbascone, Linalool, Pea Rose) arrayed between. This convergence is especially notable given the methodological shift from continuous to discrete response, and it mitigates the limitation noted after A1 regarding non-identical odor sets across experts and naïve participants.

Helmert Contrast - molecules

Comparison	Estimate	SE	df	t	p
Indole - PEA, Eugenol, Galbascone, Linalool, Galaxolide, Citronellol	0.765-	0.132	156	5.810-	001. >
PEA - Eugenol, Galbascone, Linalool, Galaxolide, Citronellol	0.474-	0.134	156	3.548-	001. >
Eugenol - Galbascone, Linalool, Galaxolide, Citronellol	0.361-	0.136	156	2.648-	0.009
Galbascone - Linalool, Galaxolide, Citronellol	0.481-	0.141	156	3.419-	001. >
Linalool - Galaxolide, Citronellol	0.444-	0.149	156	2.975-	0.003
Galaxolide - Citronellol	0.148-	0.172	156	0.859-	0.392

Table 5: Contrast

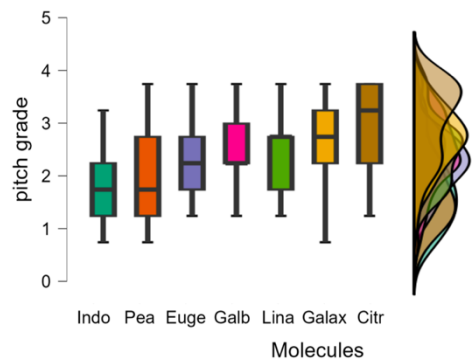

Fig 7: Frequency of pitch selection

Summary of Studies 1-3

The results reveal systematic relationships between odors and pitch, with different odors consistently associated with different frequency ranges. The marginally significant main effects suggest that both repetition and odor identity influenced pitch associations, though the effects were modest. The non-significant interaction indicates that the pattern of associations remained relatively stable across sessions.

The variability analysis provides essential insights into the nature of these associations. Some odors, particularly Indole, showed remarkably consistent pitch associations across participants and sessions. This consistency suggests stronger intrinsic cross-modal correspondences or greater perceptual stability for these odors. In contrast, the high variability observed for odors like Gallbascone and Cireonelly might reflect either weaker cross-modal associations or greater perceptual ambiguity.

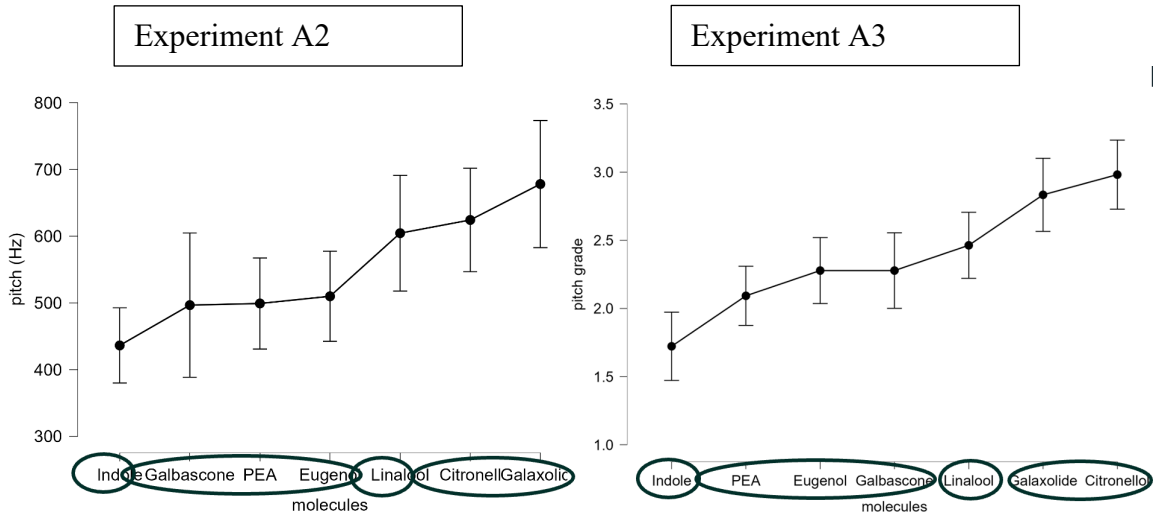

Fig 8: Consistency between-matching

Classic cross-modality matching shows that stimulus intensities from different senses can be placed on a standard psychophysical scale (e.g; Marks, 1983; Tsushima et al,. 2023; Stevens, 1959). Accordingly, in the Garner tasks, we treat pitch and odor intensity (operationalized by concentration and perceived intensity) as orthogonal candidate dimensions; interference would indicate that the two behave as an integral, standard scale rather than separable attributes.

B. Distinguishing the Two Hypotheses Through Garner's Paradigm

To investigate whether odor-pitch correspondences arise from low-level perceptual integration or higher-order cognitive processes, we employ Garner's Speeded Classification Paradigm (Garner, 1974; see also, Algom & Fitousi, 2016; Hesse & Schenk, 2013; Niv et al,. 2022). This paradigm serves as the method of choice for determining whether two stimulus dimensions are processed independently (separable dimensions) or as part of a unified perceptual whole (integral dimensions) (Algom & Fitousi, 2016).

Suppose that participants classify rectangles by height while ignoring their width. In the Baseline condition, the irrelevant dimension (width) is held constant across all trials, so that only height (long, short) varies from trial to trial. In the Filtering condition, task-irrelevant width (wide, narrow) also varies randomly from trial to trial, while height still must be classified. The task for the participant remains the same in both conditions-respond whether height is long or short. If height and width are processed independently, height performance should remain the same across Baseline and Filtering. However, if they are processed integrally, then irrelevant changes in width will impair height classification, a cost known as Garner interference (GI). Garner interference is defined as,

$$GI = MRT(F) - MRT(B)$$

where MRT(F) is the mean reaction time in the Filtering condition and MRT(B) is the mean reaction time in the Baseline condition A positive Garner interference (GI) value indicates that the irrelevant variability took a toll on performance, compromising fully selective attention to the target dimension. Conversely, no difference between Filetering and Baseline (GI ≈ 0) implies that participants successfully filtered out the irrelevant dimension, achieving perfect selective attention.

Applying Garner's logic to odor-pitch associations enables us to test whether these modalities function as separable or integral dimensions. Suppose that odor is the relevant dimension for classification. In the Baseline condition, the participant classifies odors (e.g., distinguish Odor A from Odor B), while pitch remains constant across all trials. In the Filtering condition, the participant still classifies odors, but now the irrelevant pitch dimension also varies unpredictably from trial to trial. If odor and pitch are separable, then RT for odor classification should not be affected by pitch variation, so that MRT(F) = MRT(B). However, if odor and pitch are integral, unpredictable variations in pitch will produce Garner interference, slowing down odor classification in Filtering. The Garner paradigm is illustrated in Figure 9.

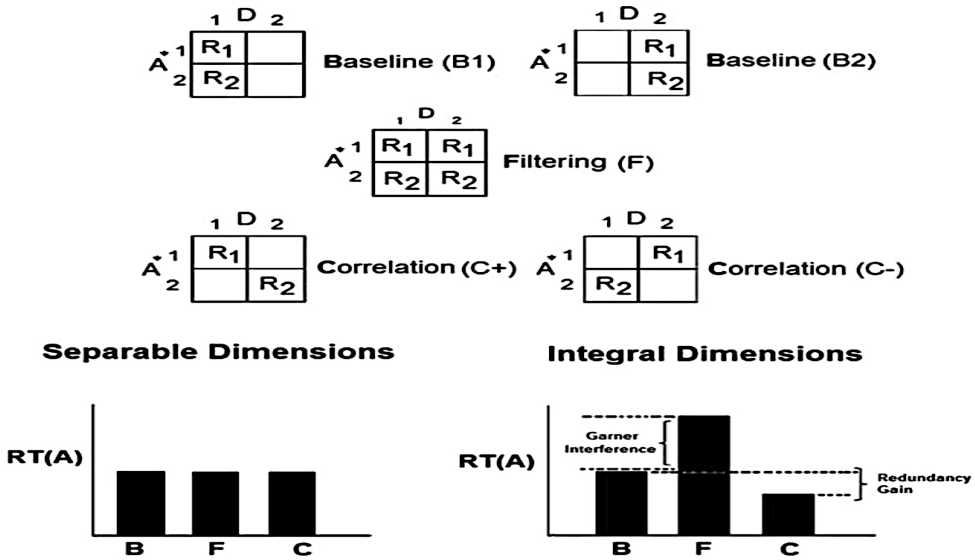

Figure 9: Schematics of Garner's speeded classification paradigm. The three basic conditions are depicted on the top. The asterisk indicates that dimension A, not dimension D, was the relevant one for responding. R1 and R2 depict the correct responses to the two values of the relevant dimension A. The Correlation conditions depict two possible ways of correspondence in the appearance over trials of the values dimensions A and D. To connect to the example of rectangles in text, dimension A is height and dimension D is width. Two prototypical outcomes, tapping integral and separable processing, are depicted at the bottom (Adapted from Algom & Fitousi, 2016). The letters B, F, and C, stand for Baseline, Filtering, and Correlation, respectively.

There is a third condition in the paradigm, the Correlation condition. It is like Filtering in that the task-irrelevant dimension also varies from trial to trial. However, in Correlation, the irrelevant dimensions vary in tandem with the target dimension. For example, if the odor is B than the pitch is high, whereas when the odor is A the pitch is low. The results in this condition can also be diagnostic with respect to separability (see Figure 9). For example, the presence of redundancy gain supports integral processing.

In summary, separable dimensions yield no Garner interference and no redundancy gain (i.e., MRT remains equivalent across conditions). In contrast, integral dimensions yield interference (Baseline < Filtering) together with redundancy gains (Correlation < Baseline).

The research plan for this section is as follows.

Study B1: Garner Classification with Odor as Target

Study B1. Naive participants will classify two odors while ignoring concurrent tones. In the Baseline block, a single mid-range tone (500 Hz) accompanies every odor. The tone is of constant frequency. In the Filtering block, the tone changes in frequency (high, low) from trial to trial in a random fashion. Mean reaction times for odor classification in both conditions will be compared. Increased classification time in the Filtering condition, i.e., the presence of GI, would indicate integral processing of odor and pitch when odor is the target dimension to respond. It shows that people are unable to ignore auditory pitch when attending to odor. Conversely, invariant response times would suggest separable processing.

Study B2: Garner Classification with Pitch as Target

Study B2 addresses the complementary question: Does odor interfere with tone classification? In Baseline, the participants will classify two tones (high vs. low) while the same odor is smelled simultaneously. In Filtering, the task is still classification of tone, but the concurrent odor varies from trial to trial in a random fashion. Again, the presence of GI - longer RT in Filtering than at Baseline - would indicate integral processing of odor and pitch with pitch as the target to respond. The presence of GI means that people cannot focus on pitch and ignore concurrently present odors.

Study B3: The Garner Paradigm with the Correlation Condition Included with Odor as Target.

Study B3 introduces correlated variation. In the Correlation condition, irrelevant pitch also varies, but unlike Filtering, it varies in tandem with the relevant odor. There are two types of correlation across trials. (1) When the relevant odor is A, the irrelevant pitch is high and when the relevant odor is B the irrelevant pitch is low; (2) When the relevant odor is A, the irrelevant pitch is low and when the relevant odor is B the irrelevant pitch is high. Due to the predictive power of the nominally irrelevant pitch, the RT to odor is expected to be short. The RT is expected to be particularly short for integral dimensions. Integral dimensions can yield redundancy gains even as compared to Baseline (Figure 3). With odor as the target dimension, large redundancy gains would support integral processing.

It is also important to observe whether there is a difference in RT between the two types of correlation. If one type yields notably faster performance than the other type, this points to a genuine association between a certain odor and a certain pitch. For example, odor A can be strongly associated with a high-pitched sound, hence yielding fast responses that are absent with low-pitched sounds.

Study B4: The Garner Paradigm with the Correlation Condition Included with Pitch as Target.

Study B4 mirrors the design of Study B3, but with pitch as the relevant (target) dimension and odor as the irrelevant dimension. The Correlation condition is identical to that described above, except that participants now classify pitch while odors vary systematically with it. As before, integral processing would be indicated by redundancy gains compared to Baseline, and potential asymmetry between congruent and incongruent odor-pitch pairings.

Theoretical Resolution: Competing Explanations for Odor-Pitch Correspondences (see possible results table in appendices)

The linguistic hypothesis suggests the odor-pitch association results from shared metaphorical or semantic concepts, such as linking "sharp" smells to high-pitched sounds due to linguistic or cultural conventions. This hypothesis does not predict the presence of Garner interference in an odor-pith speeded classification task. The sensory-synesthetic hypothesis, by contrast, suggests a shared neural infrastructure via interaction between the brain's primary olfactory and auditory sensory pathways. This suggests a necessary odor-pitch interaction to the extent that odor may automatically map onto a pitch (and vice versa). Therefore, this hypothesis predicts the presence of Garner interference in any odor-

pitch speeded classification. In other words, odor and pitch can well be separable dimensions by the linguistic perspective but they must be integral dimension by the sensory hypothesis.

In sum, our research aims to decide between these two theoretical accounts by employing Garner's speeded classification paradigm. This paradigm is a diagnostic tool to ascertain whether odor and pitch dimensions are processed independently as is possible by a purely linguistic or semantic account or integrally as would be expected from a sensory-synesthetic origin. The pattern of selective attention observed when participants classify stimuli from one modality while ignoring the other will provide critical insights into the depth and nature of these cross-modal correspondences.

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Appendices

Odors for experiment A1

Odors Expert 1

Aquaflora, Peomosa, Octyl butyrate, Syvertal, Pink Pepper oil, guaiacwood oil, Aurelione (Symrise), Ambermax 50 (Givaudan), Geraldehyde, Azarbre, Caryophyllene acetate, camphene D, pyralone, carvone D, b methyl ionone, bergamal, p-methyl acetophenone, undecanoic acid, romascone, vanilyylidene acetone.

Odors Expert 2

Phenylethyl alcohol, Eucalyptol, Benzyl acetate, Alcohol C10, Cinnamyl acetate, Acetal (acetaldehyde diethyl acetal), Bacdanol, Cedramber, D Limonene, Cuminyl alcohol, Methyl antranilate, Amber Xtreme, Dimethyl benzyl carbinol, Phenyl ethyl dimethyl carbinol, Benzyl phenylacetate, Dihydromyrcenol, Damascenone, Damascone alpha, Damascone beta, Damascone delta, Phytol, Galaxolide (50%DPG), Floralozone, Geraniol 98, Methyl heptin carbonate (Folione), Isoananat (Cyclogalbanat), Habanolide, Guayacol, Fructone, Iso E Super, Isoraldeine 95, Linalool, Javanol, Isobutavan, Lyral, Jasmolactone, Ethyl 2-methyl butyrate, Methyl pamplemousse, Raspberry ketone methyl ester, Methyl ionone gamma coeur, Melonal, Manzanate, Methyl cedryl ketone, Mayol, Magnolan.

Link to the tone generator (experiments A1-A2)

Online Tone Generator - https://www.szynalski.com/tone-generator/

Summarizes table of the predicted outcomes for each theoretical hypothesis under Garner's paradigm

Condition	Semantic-Conceptual Hypothesis	Sensory-Perceptual Hypothesis (Our Hypothesis)
Baseline	Fast	Fast (Reference)

Filtering	Minimal or no interference (participants ignore easily)	Significant, symmetrical interference (clear integration)
Correlation	Minimal or no redundancy gains	Significant and differential redundancy gains (intensity-dependent)
Condition summery	B=F=/<C	B<F<C